VIBECHECK · THE USABLE FORMAL METHODS HACKATHON · SAN FRANCISCO · NOV 2026

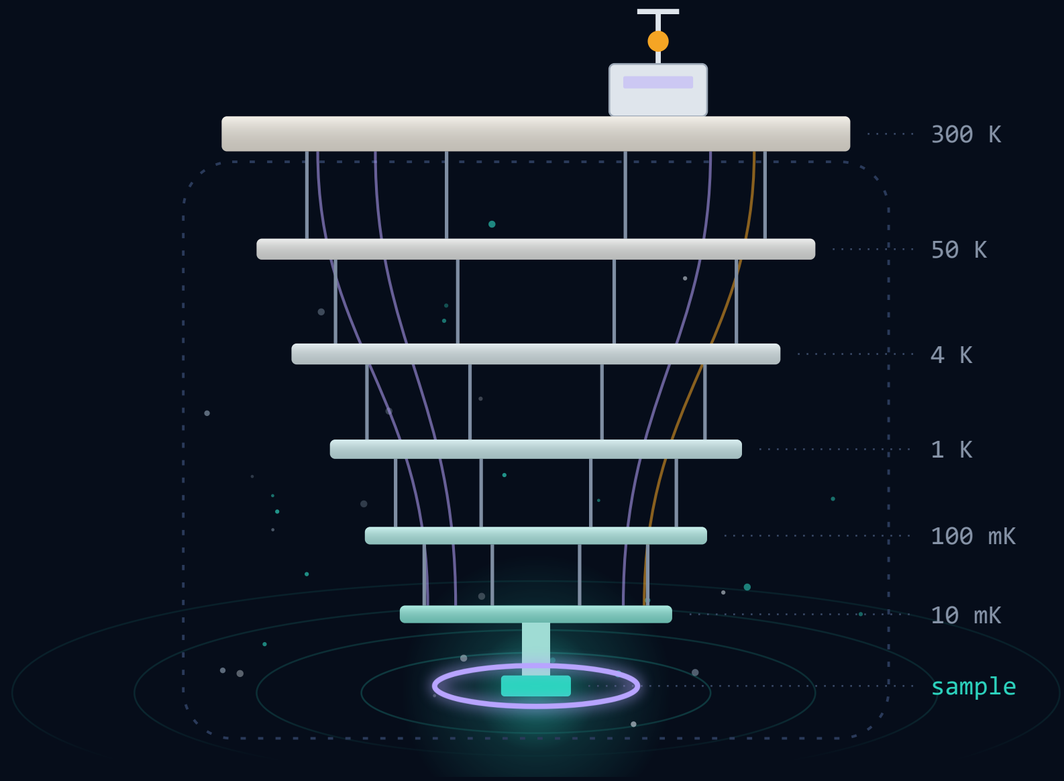
# CryoLean

## Can this cryostat do my experiment?

**Claude · Lanyon · suggest**

**Lean 4 · decides**

**Z3 · explains**

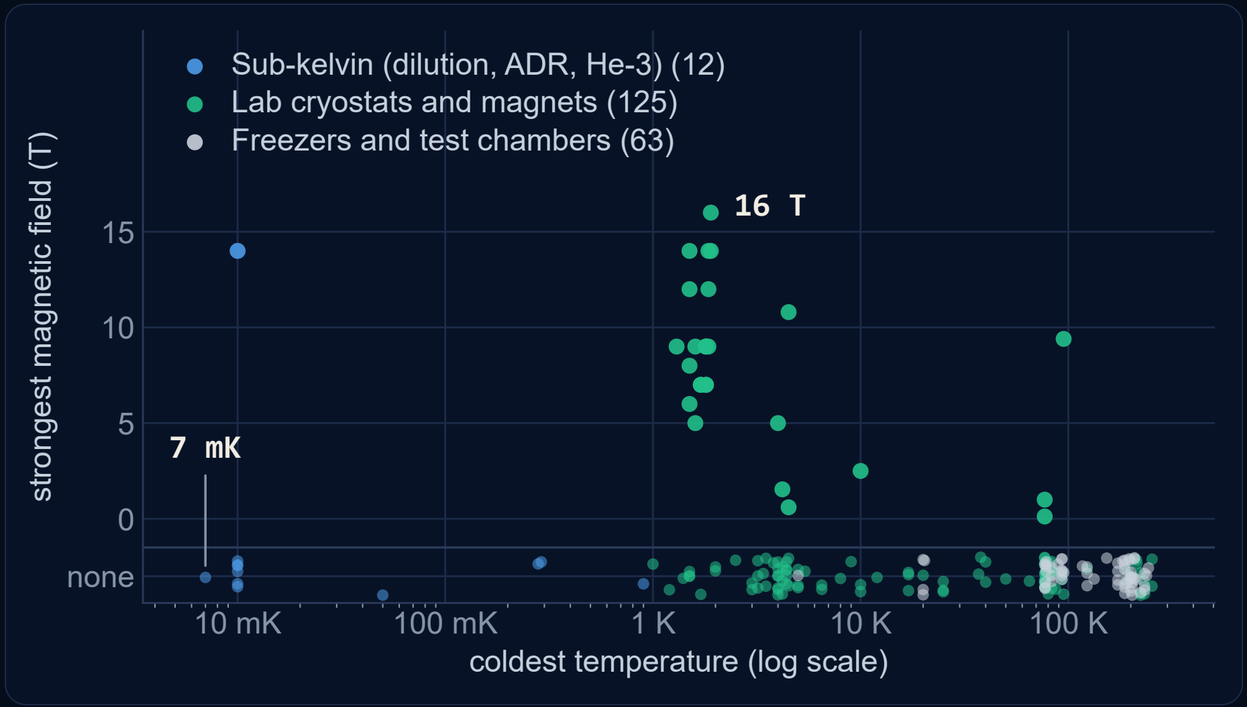



**Hanah Rahman** · Nicholas Brandenburg

hrahman12.github.io/CryoLean · github.com/hrahman12/CryoLean · generic cryostat drawn in code, no real model

/ THE PROBLEM · WHY CRYOSTATS

# A cryostat’s limits are fixed. Every experiment asks for something different.




**24 h**

to cool a Bluefors LD. A late mismatch costs that day again.

**23.5 %**

of capability entries unconfirmed (665 of 2,832).

**149** of 286

pairs a datasheet-style checker says yes to that Lean won’t sign off on.

The catalog: **220** machine configurations, **204** experiment profiles.

Chart: data/cryostats/, 200 of 220 configurations with a documented coldest temperature, field = installed magnet · data/evidence/bluefors.json · data/coverage.json · paper/data/fig_baseline_comparison.json (paper evaluation): of the 149, 148 lack trusted evidence, 1 genuinely overheats.

/ WHAT IT DOES

# Ask if an experiment fits a machine. Get the answer and the reason.

THE ASK

## Can a Bluefors LD400 run our 50-qubit wiring?

10 mK ≤ 20 mK ✓

168 ≥ 124 lines ✓

MEDIAN 138 MS PER CHECK,
INCLUDING PROCESS START

LEAN'S ANSWER

## DOES NOT MEET

Each spec passes, but together the lines put **14.592 µW** into a mixing chamber that removes **12 µW**.

## REQUIREMENTS MET

Z3 suggests 39 flux lines. Lean re-checks: **11.985 µW ≤ 12 µW**.

docs/examples/demo_b_coupled.json · paper/data/e5_heat.json · e2_latency.json · wiring from Krinner et al., *EPJ Quantum Technology* 2019.

/ PROOF IT WORKS

# Every verdict comes from a theorem.

```
theorem verdict_accept_iff (cfg : Configuration) (e : Experiment) :
    verdict cfg e = .accept ↔ AcceptSpec cfg e
```

**In words:** Lean says *requirements met* exactly when a separately written specification holds.

**128**

theorems, 0 `sorry`. 48 axiom-audited, standard axioms only.

**2,750**

fault-injected configurations. 0 defective ones accepted.

**15,795**

of 15,795 heat-budget designs: Z3 and Lean agree.

lean/CryoLean/Soundness.lean:861 · lean/audit-axioms.json · paper/data/e4_faults.json · fig_z3_lean_agreement.json. Proofs cover the modelled requirements and sourced facts, not the physical machine.